NAIMIRIS
A PROFIT DRIVEN SMART FARMING MODEL FOR SCOTCH BONNET PEPPERS

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Abstract

Small farmers who wish to grow Scotch Bonnet pepper (Nai Miris) have to take the consequential decisions before investing in land preparation, planting material, labor, irrigation and equipment. In Sri Lanka, it can be challenging to make decisions when land condition, soil evidence, water reliability, and resources and market prices are evaluated separately and informally. In this thesis, the pre-analysis and smart decision support part of the proposed profit-driven smart farming model for Scotch Bonnet peppers (NAIMIRIS) is addressed on an individual basis. The component aims to integrate land-cover data derived from satellite or aerial imagery, two guided ground photographs to capture visible land-preparation conditions, Internet of Things (IoT) observations, farmer-entered resource information, crop-reference evidence and current cost and market data. The design is based on a staged evidence model, which is a model that distinguishes model predictions, sensor measurements, farmer inputs, sourced values, proxies and assumptions. The limitation is assessed based on the FAO approach, and there is a further limitation due to soil or method of cultivation, budget, availability of labor and water. Financial output should be designed to provide ranges of income and risk probability, not a guaranteed amount of income. The research design is design centered and includes a reproducible process, evidence quality gates, assessment criteria, privacy protection and a modular structure. The main contribution is a clear, transparent pre-cultivation decision framework in a niche crop and smallholder setting that explicitly deals with missing evidence and uncertainty. Before it can be determined whether the documented design will be predictive and/or economically appropriate, data sets need to be validated and agronomic adjustments made, as well as field evaluation.

Summary: This study aimed to develop a land suitability assessment system for the cultivation of Scotch Bonnet pepper using a semantic segmentation approach. This study was conducted to develop a land suitability assessment system for the cultivation of Scotch Bonnet pepper based on semantic segmentation approach.

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ABBREVIATIONS

AI	Artificial intelligence
API	Application programming interface
DSS	Decision support system
EC	Electrical conductivity
FAO	Food and Agriculture Organization
IoT	Internet of Things
IoU	Intersection over union
MAE	Mean absolute error
OEM	OpenEarthMap
ROI	Region of interest in imagery; return on investment in finance
RGB	Red green blue
VWC	Volumetric water content

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1 INTRODUCTION

1.1 Overall project context

While agriculture continues to be a key activity in Sri Lanka and crops are a choice based on climate and crop type, farmers' decisions do not just depend on the climate and seed they choose. The land should be physically accessible, the soil and drainage properties should be suitable for the crop, water must be available in adequate supply and labour and capital must be available. Failure to take any of these factors into account can result in unnecessary costs or a plan that isn't achieved. A poor pre-cultivation choice can have serious implications for a smallholder as it could be a significant proportion of household or business resources [2].

NAIMIRIS is a project of group research, with the name A Profit-Driven Smart Farming Model for Scotch Bonnet Peppers. It is geared towards the support of small-scale farmers and potential investors in Sri Lankan's Wet Zone. It encompasses pre-analysis and decision support, technology-assisted cultivation guidance, a digital marketplace and price intelligence, and post-harvest value addition. It is a first component of this kind of decision which is addressed in this individual thesis, specifically the question of whether a planned plot may be used before large sums of money are invested in the cultivation and, if so, how it can be.

Scotch Bonnet is a high heat Capsicum pepper with a distinctive flavor and aroma. The general guidance for capsicum or chilli, however, cannot be assumed to be validated general guidance for all varieties of Scotch Bonnet. The Sri Lanka Department of Agriculture provides the soil and cultivation specifications for capsicum (Solanum annum L.), which includes well-drained fertile loam soil and preferred soil pH range but such information is a crop-family reference that needs to be validated locally for the selected variety of Nai Miris capsicum, Solanum annum L. [1]. Therefore, for general recommendations on chilli/capsicum, NAIMIRIS considers these as proxy evidence unless variety level evidence is available.

Digital decision support can enable structuring of information that would otherwise be spread across the maps, farmer knowledge, sensor readings, cost records, and market observations. However, the use of digital tools should not supplant the need for sound evidence. The image classifier cannot determine soil pH, the soil-moisture sensor

cannot determine soil texture and the retail price cannot be assumed to be the price a farmer will receive. The system is therefore geared towards identifying evidence used, uncertainty and what further evidence is required before a recommendation is reached [3].

1.2 Individual component context

The single component is the NAIMIRIS Pre-Analysis and Smart Decision Support Stage. It is used to assist a farmer to assess a proposed plot and formulate a viable farming plan prior to planting. The component is not designed to be a stand-alone agronomist or sales application. Rather, it builds on multiple evidence sources to create an explicit and unambiguous decision workflow and presents constraints to the user.

There are two functions for image analysis in the design. Model 1 uses a satellite (or any other kind of appropriate overhead image) to estimate the distribution of classes of land cover within a region of interest defined by the farmer. Agriculture, barren/open land, buildings, roads/access and water are among the list of candidate classes, along with forest/trees, developed/other and unknown classes. The final FAO suitability class and the final feasible plant count are not given in Model 1; it only gives the land evidence and an initial range of planting capacity. Model 2 is being used to analyse only two guided ground photographs from opposite sides of the plot. It detects any visible preparation conditions like high grass, bushes, bare land, trees, mud, standing water and debris. Perspective photographs are not reliable for determining the exact proportion of the area damaged by each condition, and preparation quantities and costs need to be determined by the farmer after confirming the proportion damaged by each condition.

The component also takes four IoT observations: soil moisture, soil temperature, air temperature and air relative humidity. These values are used in environmental observations, but do not reflect the soil texture, pH or electrical conductivity (EC) [4]. The evidence of soil should thus not be inferred from the four sensor variables, but should be left out until it is found. The design includes the following features: The suitability is assessed in stages according to limiting factors, and the current suitability is differentiated from the potential suitability following justified improvements [2].

The financial decision engine integrates planned plant count, labour availability, owned equipment, purchase/rental options, preparation activities, capital expenditure, operating expenditure and monthly farm-gate or contract market price data. If there is no price or yield data available for a particular crop, a proxy may only be displayed explicitly with a lowered level of confidence and a warning [5]. The result is an estimate in order to make decisions—not a guarantee of farm income.

1.3 Problem statement

Before investing in Scotch Bonnet, small farmers interested in growing it may not have a convenient way of integrating information about land suitability, land use evidence, water availability, manpower and machinery, budget constraints, and potential farm-gate returns. These factors are interdependent: an area that may look like it is available could need a lot of money to clear; an area that might seem suitable could be greater than the farmer's water or labour capacity; an estimate that seems to be profitable may be incorrect if based on retail prices, not verified yield assumptions, or incomplete cost data. There are existing technical approaches that can assist with individual activities (e.g., land-cover classification, crop modelling, sensor monitoring), but outputs of these approaches do not necessarily result in a farm-specific decision that can be tracked back [4]. The research problem, then, is how to design a pre-cultivation decision-support component that is transparent, incorporates heterogeneous evidence, explicitly addresses missing information and uncertainty, and recommends a physically/ economically feasible Scotch Bonnet cultivation plan without being overly prescriptive with respect to what the data can support.

1.4 Research gap

Three gaps that are related motivate the component. The first is that land-cover classification and the recognition of ground-condition are two different tasks. Although overhead imagery can be used to estimate the distribution of land-cover classes, and ground photographs can provide information that can be used to identify preparation needs, neither alone is adequate to estimate the full cost or agronomic

suitability of a plot. Secondly, the suitability of an agronomic application relies on factors that cannot be taken from images or the existing IoT device [2]. Lab results, verified records, farmer confirmation or specialist assessment might be required for soil texture, pH, drainage, and water reliability and crop-specific requirements. Third, a feasible cultivation plan might not be technically feasible due to budget, labor, water or market requirements. The backend specification remedies these shortcomings by introducing explicit Evidence categories, required decision gates and a required final calculation of feasible number of plants.

The crop specifies is also significant as the general recommendations for capsicum/chilli might not have been tested for all the varieties of the species grown in Sri Lanka [1]. The proposed contribution is not a claim of a lack of related system. Instead, the use of land evidence, visible preparation conditions, limited IoT measurements, agronomic evidence, resource limitations, and risk-aware financial estimation is combined into a single, traceable, and pre-cultivation workflow that suits the needs of Nai Miris. The novelty and the utility must be evaluated at the end in comparison, implementation and field validation.

1.5 Aim and objectives

1.5.1 Main aim

To develop a clear, evidence-based pre-cultivation decision support system that assesses land suitability, resource limitations and then estimates the likely options and return for Scotch Bonnet cultivation and return by smallholders in Sri Lanka.

1.5.2 Specific objectives

1. To develop a land-analysis workflow to estimate proportions of land cover and preliminary planting capacity of a plot boundary and suitable satellite/aerial imagery.
2. Develop a two-photo land-preparation analysis workflow which incorporates farmer confirmation to generate preparation tasks for visible conditions.
3. Identify an evidence pipeline for soil moisture, soil temperature, air temperature and relative humidity for IoT applications, covering calibration, quality checks and stale data management.

4. Elaborate a staged land suitability decision process using an approach similar to the FAO limiting factor approach and providing a provisional classification where needed, when necessary evidence is lacking.
5. Apply land, soil/method, budget, labor and water limitations to determine a final viable number of plants and compare plantings options.
6. Create a clear financial estimation procedure, which distinguishes sourced data from proxies and assumptions and provides ranges and probabilities of risk instead of guarantees of return.
7. Provide criteria for evaluation, data provenance, privacy protection and reproducibility for future implementation and validation.

1.6 Research questions or hypotheses

The following research questions are used to guide the study:

RQ1: What is the potential of using overhead imagery and farmer-defined ROI to estimate the initial usable area and planting capacity while showing uncertainty?

RQ2: What are the possible ways to use two guided ground photographs, with farmer verification, to estimate land-preparation requirements without requiring survey-grade area measurement?

RQ3: What should be the influence of incomplete evidence (IoT, soil, crop-reference, and water) on the land-suitability decision of the system?

RQ4: What is the feasibility of the plant count when considering physical capacity, soil or cultivation method, budget, labor and water constraints?

RQ5: How to communicate financial uncertainty and source provenance in a way that helps instead of hinders farmer and specialist judgement? These are not yet hypotheses but research questions at this stage. When datasets, reference labels, implementation and evaluation protocol are established, quantitative hypotheses should be defined.

1.7 Scope, limitations and assumptions

1.7.1 Scope

- The pre-analysis and smart decision-support part of Scotch Bonnet/Nai Miris cultivation is called the individual scope.
- Land input may be provided as polygons, using a map, or as an image uploaded with a hand-drawn boundary.
- The overhead image model estimates land-cover classes and preliminary physical capacity; while the ground-image model employs exactly two instructed photographs.
- So far, the current specification only specifies integration with soil moisture, soil temperature, air temperature and air relative humidity of the IoT.
- The decision logic takes into account evidence completeness, land suitability, cultivation method, planting capacity, budget, labour, water, preparation and financial uncertainty.
- The outputs are: explanations, warnings, alternatives and a report of sources, model versions and assumptions.

1.7.2 Limitations

There are some important limitations to the proposed design. Overhead imagery may be dated, cloudy, low resolution or be inconsistent with plot conditions. The models developed using the public land-cover data might not be fully representative of the Sri Lankan small farms, local vegetation or mixed land-cover scene. There is only partial reconstruction of the 3D plot and an exact determination of the affected area can only be made from two ground photographs. Soil texture, pH or EC cannot be determined by four IoT variables. Recommendations for general chilli/capsicum use are not the same as locally proven Scotch Bonnet agronomy. Market records may be lacking or provide retail information instead of farm-gate information. The yield, survival, grade, labor productivity and selling price may differ according to the season and farm practice. Such limitations require uncertainty to be presented and manual/expert review when necessary [1].

1.7.3 Assumptions

The design makes some assumptions about the user being able to submit a valid plot boundary or land image, answer simple questions regarding budget and resources and validate visible preparation conditions. It makes the assumption that appropriate imagery and legally usable source data is available. It also makes assumptions that unit cost, crop space, labour coefficient and price records can be versioned and/or modified. Such assumptions need to be checked in implementation and field testing.

1.8 Individual contribution

The pre-analysis and smart decision-support stage is the individual contribution for student IT22271464. This involves specifying the evidence model, land-cover and two-photo analysis contracts, the IoT data-quality and soil-evidence limitations, designing the FAO-style suitability gate, including plant capacity with budget, labour, water limitations, and defining the financial-risk and reporting rules. The contribution is separated from other group elements that provide A-Z guidance to cultivation, digital marketplace and price intelligence, and post-harvest value addition. This thesis is meant for presentation of the design contribution; the following chapters on the implementation artefacts and evaluation results should only be given if supported by concrete evidence.

1.9 Thesis structure

Chapter 1 introduces the project, individual component, problem, research gap, aim, objectives, research questions and scope and individual contribution. Chapter 2 examines technical and domain foundations, related research, existing datasets, and existing techniques and the gap is synthesized into a conceptual framework. The design-oriented research methodology, research materials, procedure, evaluation measures, data analysis, ethics and reproducibility are described in Chapter 3. Requirements, component architecture, design alternatives, detailed data and decision flows, and standards for professional practice are presented in Chapter 4. The issue of implementation, integration and testing is deliberately omitted from this document (Chapter 5). Supplementary material is provided in Appendix A (requirements

traceability matrix), Appendix B (summary of formulas and decision rules) and Appendix C (academic integrity and AI use disclosure).

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

This chapter does not constitute a full systematic literature review but is based on a focused review of the literature which was driven by the need to address a set of problems. The review scope will be limited to the following sources: land evaluation, remote-sensing land-cover segmentation, overhead-image acquisition, ground-image condition recognition, financial uncertainty and responsible AI, and the individual components will be evaluated. The source base includes the proposed topic, the final backend specification, official agricultural guidelines from Sri Lanka, the Food and Agriculture Organization (FAO), dataset papers and documentation from image-providing organisms, and an AI Risk-Management framework [4].The number of sources chosen was based on the relevance to design decisions, not to establish the effectiveness of the system. The selection of candidate training or benchmarking material is not an indication of good performance of a model trained on these datasets on Nai Miris farms. Official guidance is applied as reference, however, it is noted that it is applied as a proxy for Scotch Bonnet if variety specific validation is not available. The specification for the back-end is considered to be the design authority, not as a piece of evidence that predictions have already been validated [1]. The review was further extended on 10 October 2026 using dedicated web search on the following: LoveDA, OpenEarthMap, FAO Land suitability, Farm Monte Carlo simulation, design science research, DSSAT pepper, GeoJSON, candidate segmentation architectures. Sources supported a design decision were retained, as can be cross referenced with an original paper, official institutional publication or provider documentation. This is a focused review and not a systematic review; there is no screening count and there is no comprehensive product ranking made.

2.2 Domain and theoretical foundations

2.2.1 Land evaluation and limiting factors

Land evaluation involves comparing the characteristics and qualities of the land to the requirements of land use to determine suitability of the land for a particular use. The FAO framework has a classification of suitable and not-suitable orders, and the classes

S1 (highly suitable), S2 (moderately suitable), S3 (marginally suitable), N1 (currently not suitable) and N2 (permanently not suitable). This classification is use specific and a limiting factor that is very serious may be the reason to decide the overall classification [2]. This is suitable for NAIMIRIS as a good image classification cannot offset a critical restriction like poor drainage, poor soil evidence, poor water or an economically uncorrectable restriction. The framework also does not ignore current suitability but rather considers it in relation to potential suitability following improvement. NAIMIRIS follows this distinction: The plan and its costs can be altered if there are drainage improvements, organic-matter amendments, or if a different cultivation method should be used, but not if the original limitation is disguised or the land gets automatically reclassified without proof. If there is missing or dated evidence, then it should be returned with "PROVISIONAL – NOT ASSIGNED", missing evidence and safe interim actions. This will prevent that an unsupported suitability class be concluded as an agronomic conclusion [2].

2.2.2 Remote sensing and semantic segmentation

The observation of the earth from space remote sensing can be used to assist land cover interpretation. Semantic segmentation provides a class for each pixel that can estimate the percentage of a chosen region that falls into the categories of agriculture, barren land, buildings, roads, water, trees, and other/unknown cover. The LoveDA is a remote-sensing land-cover dataset, which is used to deal with domain-adaptive semantic segmentation and contains both urban and rural images [4]. OpenEarthMap offers high resolution aerial and satellite images which are manually annotated and available in different areas and for eight land-cover classes [6]. These datasets are used as candidate technical foundations for experimentation but can have variations in geography, image resolution, class definitions and local land use patterns, which pose a risk of domain shift. The class probability of a model is not necessarily the same as the ground truth. As a result, the following class probabilities, confidence information, model version, image source, capture date, and resolution are preserved in the design. The region of interest selected is stored separately as a binary mask. The segmentation model is given a rectangular RGB crop, and outside the polygon on the image is not considered when aggregating areas. Furthermore, this design does not require the use

of an irregular transparent image as the model input and maintains the separation between the image crop and the actual portion to be analyzed [4].

2.2.3 Ground-image analysis for preparation conditions

Both overhead and ground photos offer evidence. Overhead imagery is helpful for the spatial extent and land-cover distribution, while ground-level images might show visible vegetation, mud, standing water or other debris that could be of interest in preparation. The back-end specification only allows for two guided photographs from two opposite sides of the plot. Both the images are analyzed in terms of blur, exposure, viewpoint repetition and usable portion of the scene. Only retake one image if it fails to pass quality checks. The two-photo design is intentionally a conservative one. Does not claim to recreate an entire story or work out the actual areas of interest from perspective views. The model suggests condition classes and severity, and the farmer validates a fraction of affectedness. The confirmed fraction and the candidate area derived from the overhead is then added to the unit-cost and labor coefficients sourced from the preparation to determine the preparation ranges. This way, there is less false precision and the user becomes a more explicit part of the decision-making process [3].

2.2.4 IoT observations and evidence completeness

Sensor data from the IoT can offer multiple environmental observations that may not be possible to see with one manual inspection. Soil moisture, soil temperature, air temperature and air relative humidity are the four readings in the current design. Only if the device is authenticated, the timing and sequencing are handled, the calibration metadata is provided, plausible range checks are performed, spikes are detected, and stale data is detected and stored along with the calibrated data can a reliable use be made of it. The missing values should not be replaced with zero as this would be read as an observation, not absence of evidence. The four sensor variables cannot define the ratio of sand, silt and clay and thus can produce no soil texture. They are not capable of setting pH or EC also. This property needs an appropriate laboratory result, suitable sensor, verified soil map or a documented manual assessment. NAIMIRIS thus considers the IoT readings a single evidence category and not a complete soil test.

2.2.5 Crop-reference evidence and local applicability

The Sri Lanka Department of Agriculture provides a variety-specific spacing and cultivation guidance along with the identification of suitable soil type as well as a preferred range of soil pH for capsicum cultivation [1]. These values can be used in the design of evidence fields and provisional checks; however, these values should never be used as universal values for every cultivar of Scotch Bonnet. The specification in the backend must then include crop, variety, source and version along with the spacing or agronomic thresholds. All proxy guidance should be marked and the confidence level reduced until local evidence is available.

2.2.6 Financial decision support under uncertainty

The feasibility of financial dollars is not just based on one revenue estimate. Land preparation, irrigation, equipment, seedling, labor and operating expenses have to be matched with reasonable yield and price expectations. In the field of farm investment and profitability, the FAO's farm-management material defines Monte Carlo simulation as a good tool to analyse risky farm investment and profitability decisions [5]. In NAIMIRIS, simulation is suggested as a means of capturing uncertainty in yield, survival, quality, price, labour productivity and cost, and as a way of reporting a range of possible outcomes instead of one guaranteed outcome. The financial engine should also maintain the difference between farm-gate/ contract/ wholesale prices and retail prices. Consumer demand may be evident in a retail listing, but cannot be equated to how much a farmer can receive. If there is no price evidence for a particular crop, a proxy crop (such as green chilli or capsicum) may be used provided that it is clearly labelled, traceable and lower confidence. When estimating a break-even size and return on investment range, assumptions must be used and it should be noted that the income predicted is not certain [5].

2.3 Related research

2.3.1 Land-cover datasets and domain adaptation

LoveDA and OpenEarthMap are all relevant since LoveDA offers a land-cover segmentation research dataset with urban/rural domain variation and OpenEarthMap offers global high resolution annotations across a large variety of regions [31]. They

have their own class taxonomies which can be used as a starting point for mapping labels from the sources to NAIMIRIS classes. The source classes are not necessarily aligned with the categories needed for farmer decision making, however. For instance, “other” or “background” classes should not be assumed to be suitable for planting and “barren” does not mean "agronomically barren". So, a mapping layer and an unknown/low-confidence category is therefore required. The problem of transfer to the small plots in Sri Lanka is a key research challenge. Examples of local farms, vegetation, shadows, mixed crops, seasonal conditions and image providers do not necessarily represent the range of conditions found in the dataset for multiple countries. A valid evaluation should contain local reference annotations, and should provide information on the performance of the classes as well as the overall performance. It should also be investigated whether or not boundary errors affect the usable area and the planting capacity significantly, as estimated [4].

2.3.2 Image analysis combined with user confirmation

We need to use image analysis and user confirmation in conjunction with each other. But an automated system that does not consider the uncertainty in the visual evidence can turn it into unfairly exact costs. NAIMIRIS, on the other hand, integrates model results with farmer's confirmation in the decision points. In the case of Model 2, the model will predict conditions based on what is visible, and the farmer will validate the amount of area that is affected. User can check and modify the selected boundary and modify the planned plant count for Model 1. These interactions help to maintain traceability, and also minimize the risk of a classifier's confidence score being confused with a survey measurement [3]. However, there is a trade-off as user confirmation adds variability and can be a bit confusing. Future evaluation will involve measuring consistency of different users' interpretations of bands of areas affected, frequency of override of proposed plant counts, and proper communication of limitations. Human confirmation is not assumed to be a correct evidence source, it is another source of evidence which must be recorded [3].

2.3.3 Land suitability and economic feasibility

The FAO framework offers a conceptually based approach for matching the land quality into a specified use [2]. NAIMIRIS is expanded to a decision workflow, which takes into account not only agronomic suitability but also the affordability and labor, budget and water availability of the farm. Final plant count is the lowest of the farmer's target plant count and the land, soil/cultivation system, budget, labor and water capacities. This clearly differentiates between the physical ability and viable operation. The extension must be read carefully: A financial constraint must not be hidden behind a soil constraint and an alternative like bag cultivation must be treated as a different cultivation system and must be assessed assuming other cost and capacity figures. This is a method whose outcome is not to alter the original land condition (direct-soil cultivation) [2].

2.3.4 Risk-aware estimates and transparent evidence

Monte Carlo simulation is a way of simulating uncertainty by sampling multiple realizations from known distribution of variables [5]. For NAIMIRIS, inputs can be Yield, Survival rate, Saleable grade, Farm-gate price, Preparation cost, Labor hours and Input prices. Metrics that can be supported from the resulting distribution include median profit, probability of loss and probability of exceeding the budget. This kind of outputs only have relevance if they make use of evidence or if they are marked as such. Even if a large number of simulations is used, it can make up for poor and/or biased input data. The evidence model in the back-end specification classifies the data as model output (M), IoT measurement (I), farmer input (F), sourced value (S), proxy (P) or assumption (A). The following is an implementation of a provenance mechanism. The source, date, units, region, channel or model version (as appropriate) of each value should be maintained. A report must provide information on evidence completeness as well as uncertainty; evidence of the soil and drainage evidence is essential even if the model is highly confident [3].

2.3.5 Candidate segmentation architectures

U-Net is a contracting path network along with an expanding path network, which is connected with features for the purpose of localization [7]. DeepLabv3+ uses a decoder designed to refine the object boundaries in addition to multiscale context [8]. Both are good overhead land segmentations comparison candidates. Here they are not specified as the NAIMIRIS architecture trained: they are confirmed by the specification provided, but the shape of the input tensor, the loss function, the decoder and the encoder are not defined. All class mapping, geographic split, preprocessing, evaluation mask and training budget are assumed to be constant to make a fair comparison. Report per class IoU, usable-area error and inference cost are all together. Although the best performing model in terms of mean IoU, it might still be inappropriate because of the number of buildings and agricultural areas misclassified. In contrast to model 1, multi-label outputs are needed in model 2 as grass, water and debris can all co-occur in a single image; if the same softmax class is used, it will ignore the other valid co-occurring conditions [7].

2.4 Existing systems, products and techniques

The reviewed sources are not intended to replace NAIMIRIS workflow, but rather are intended to be used as enabling data, technical services and agronomic frameworks. They are outlined below to highlight their relevance and limitations. [1]
Table 2.1 lists some of the related techniques and their applicability.

Table 2-1 Related techniques and applicability

System / technique	Primary relevance	Limitation for this component	NAIMIRIS design response
LoveDA dataset [4]	Remote-sensing land-cover segmentation and urban/rural domain variation.	Not a ready-made Scotch Bonnet suitability model; geographic and class-domain shift remains.	Use as candidate training/benchmark data with local validation and explicit class mapping.
OpenEarthMap [6]	High-resolution global land-cover annotations from aerial and satellite imagery.	Global coverage does not guarantee local small-farm performance or agronomic suitability.	Retain unknown classes, imagery metadata and local validation requirements.

System / technique	Primary relevance	Limitation for this component	NAIMIRIS design response
FAO land-evaluation framework [2]	Suitability classes and limiting-factor reasoning.	Requires crop-specific thresholds and adequate land/resource evidence.	Use a provisional gate and keep current versus potential suitability distinct.
Sri Lanka DOA capsicum guidance [1]	Local crop-family reference for soil, pH, spacing and management.	Not necessarily validated for every Scotch Bonnet variety.	Store source and variety version; label as proxy where appropriate.
Satellite imagery services [9]	Provide mechanisms to obtain imagery or export selected regions.	Licensing, spatial resolution, image age, clouds and API constraints affect use.	Use permitted sources and retain capture date, resolution and provider metadata.
IoT sensor monitoring	Repeated readings for moisture and local environmental conditions.	Four current readings cannot establish soil texture, pH or EC.	Treat sensors as partial evidence and request laboratory or verified soil evidence.
Monte Carlo simulation [5]	Risk-aware analysis of uncertain farm investment outcomes.	Results depend on credible distributions and local calibration data.	Report ranges and probabilities, label assumptions and avoid guaranteed income claims.

Crop information, disease identification, farm management or market connectivity could be offered by commercial farming platforms or by applications that would provide farm advice. The present review is not an exhaustive, independently-tested feature-by-feature comparison of all such products. NAIMIRIS is therefore not claiming to be the first or only system to offer advice for smart farming. It proposes a distinction of a narrow scope pre-cultivation workflow for Nai Miris, combined with gates related to evidence, feasibility of resources and risk-aware financial estimation. CROPGRO-pepper is included in DSSAT for bell pepper [10]. This provides a valid crop-simulation comparator, but not a comprehensive simulation preparation-and-budget process for local Scotch Bonnet calibration. The current design does not assert that a calibrated crop model is integrated; it will be desirable to include a calibrated crop model in the future.

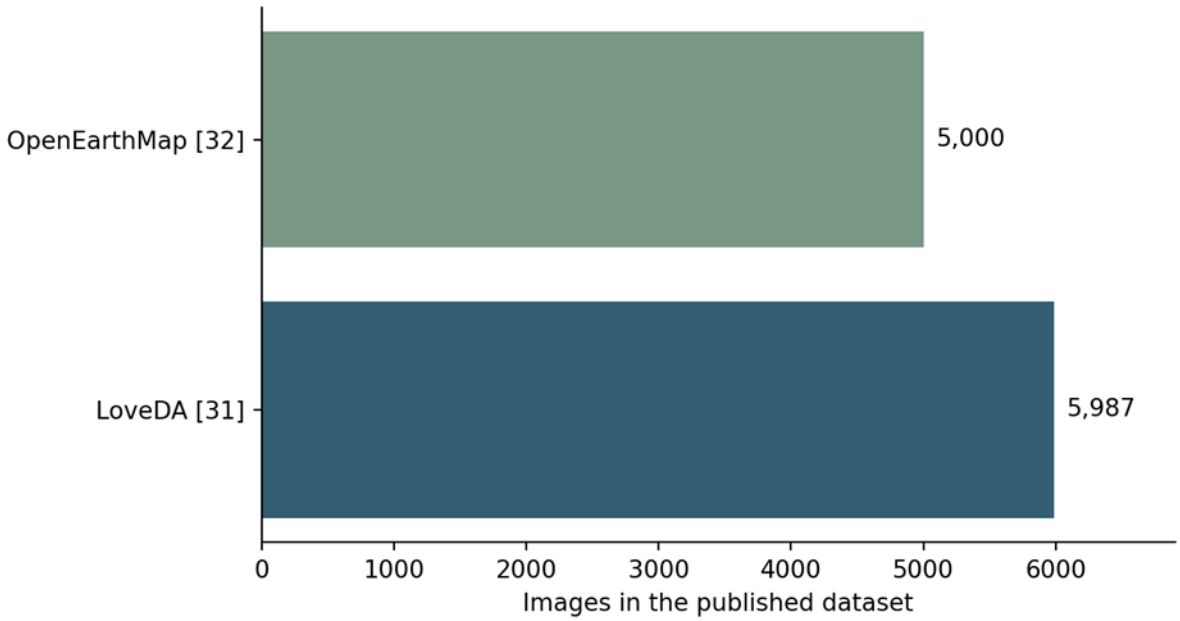


Figure 2-1 Publish dataset

Figure 2.1 Published image counts for candidate overhead datasets [4]

The figure 2.1 shows a comparison of the number of datasets available when published, not the number of images downloaded, mapped or used for NAIMIRIS training. The latter must be based on a saved training manifest. Local suitability or model quality is not a measure of dataset size. [4]

2.5 Gap synthesis and conceptual framework

A literature review and design review showed that there is no single evidence source that is enough for a defensible pre-cultivation decision. Overhead imagery is used to help with area and land-cover estimates; ground photographs are used to help with visible preparation-condition estimates; IoT readings are for time-stamped environmental observations; crop-reference sources are for provisional agronomic checks; and farmer inputs are for budget, labor, equipment and water availability. The proposed plan is then translated into financial records to show a financial range of costs and returns. The sources must be linked together but they cannot lose their distinct evidence status [4]. The conceptual framework consists of a series of decisions which are gated. Plot boundary and imagery is validated first. Secondly, Model 1 is estimating distribution of land cover and physical capacity. Thirdly, Model 2 was

designed to recognize preparation conditions visible from only two quality checked images, and to have the farmer confirm fractions affected. Fourth, checks to see if IoT and external soil evidence are complete. Fifth, suitability is assessed based on the limiting factors: PROVISIONAL – NOT ASSIGNED if necessary evidence is lacking. Sixth, farmer planned planting count is validated against farmer physical, soil/method, budget, labour and water capacities. Last but not least, risk-aware financial estimates are filled with sourced costs and evidence of market price. The report documents the source, date, model version, confidence and assumptions of each significant result [2].

Two ground photographs	Visible condition classes and severity	Preparation tasks and cost ranges	Retake failed image; farmer confirms affected fraction; no exact-area claim.
IoT + soil/crop evidence	Time-series sensor summaries and evidence completeness	Current/potential suitability assessment	Missing pH, texture, drainage or water evidence prevents final class assignment.
Farmer resources	Budget, workers, working hours, owned equipment and water details	Resource and cultivation-method feasibility	Validate units; preserve user input and corrections.
Cost and market evidence	Low/base/high cost and revenue assumptions	Break-even and risk-aware profitability	Use farmer revenue channels, not retail prices; label proxies and assumptions.
Integrated decision engine	Feasible plant-count range, alternatives, warnings and report	Pre-cultivation plan choice	Final count is limited by land, soil/method, budget, labor and water.

Table 2-2 Evidence flow and conceptual framework

The conceptual contribution is not about predicting the success of a farm based on which AI model is used, but rather it is about an evidence-based decision-making process. The way forward is to make uncertainty actionable—when there is uncertainty, ask for evidence; when it is limited, record an alternative; and when it is short, put in place a smaller plan or another way of working. The methodology detailed in Chapter 3 and detailed architecture detailed in Chapter 4 are each based on this framework [3].

3 RESEARCH METHODOLOGY

3.1 Research design

Design oriented research method was used in the study. At its core, there is a pre-cultivation decision support system that combines land evidence derived from images, land conditions observed with the naked eye, data from IOT sensors, farmer information, crop references and financial assumptions. The research steps include defining the problem, requirements traced from the literature, conceptual design and data-flow design, and a planned protocol for evaluating the decision rules, which are specified. This is appropriate since the research question is directed toward how diverse evidence can be aggregated into a traceable decision process and not at testing a single isolated algorithm [11]. The work consists of four methodological activities. First, the requirements and limitations are derived from the proposed and the consolidated backend specification. Second, external sources are referenced to substantiate the land evaluation, candidate datasets, handling of crop references, acquisition of imagery and responsible AI principles. Thirdly, the component is defined in terms of inputs, quality gates, calculations, outputs and failure states. Fourth, a plan is established for evaluating the strategy in the future. In this report no claim is made for completed field testing, measured model accuracy or verified economic benefit [4]. The system implements a hybrid decision approach, where classification of images is done by machine-learning models and explicit constraints and evidence gates are applied in a rule-based logic. This hybrid will be selected for traceability. A classifier can recommend a land cover class, but alone cannot determine suitability of the land cover or economic viability of the project. [3] The study is designed in accordance to the 6-activity Design Science Research methodology proposed by Peffers et al. [11] which includes problem identification, objectives, design and development, demonstration, evaluation and communication. Chapters 1-4 set up the problem, objectives, design and evaluation protocol. Later, there needs to be actual artefacts and records to back up demonstration and evaluation.

Table 3-1 Research question and evaluation traceability

Question	Comparison or observation	Primary evidence
RQ1 Land and capacity	Segmentation against held-out manual masks; capacity against reference layouts	Per-class IoU; area MAE; blocked-land false positives
RQ2 Two-photo preparation	Photo A only, Photo B only, pair and farmer-confirmed pair	Macro F1; condition recall; task-quantity error
RQ3 Incomplete evidence	Complete cases versus missing, stale and contradictory cases	Unsupported final-class count; expert agreement
RQ4 Resource feasibility	Land-only count versus constrained count	Constraint violations; binding factor correctness
RQ5 Financial communication	Point estimate versus ranges on the same inputs	Interval coverage when outcomes exist; warning comprehension

3.2 Data, participants and study materials

3.2.1 Study materials

These are the types of study materials that are incorporated in the design:

- Information including plot-boundary geometry, capture date, spatial resolution, source/license metadata and overhead imagery (satellite, aerial or appropriate uploaded imagery).
- Data sets: Land-cover data sets - LoveDA and OpenEarthMap - as potential sources for model development and/or benchmarking (to be verified by licensing and class mapping and local validation) [4].
- Ground photographs: Only two guided photographs were taken from opposite sides of the plot selected (image quality checked).
- IoT observations: timestamped soil moisture, soil temperature, air temperature and air relative humidity, with raw, calibrated values, units and calibration version.
- Agronomic evidence: official crop family guidance, local plant needs for Scotch Bonnet (where available), soil lab data, soil maps, observation of drainage and water. [1]
- Farmer-entered information: land size/boundaries, budget, planned plants, available workers and working hours, owned equipment, water source / reliability and preferred cultivation method.

- Cost & market evidence: outdated local unit costs, labour prices, farm-gate/contract/wholesale prices, crop yield and quality records, and proxy/assumption values that are clearly marked.

3.2.2 Participants and validation stakeholders

Future testing of the proposed system might include small-scale farmers who are currently growing Nai Miris or who are thinking about starting to grow it. It could also involve professionals who know a lot about Capsicum farming and technical evaluators who can check how well the system works and review the results it produces.

Even though a sample questionnaire for farmers is included in the proposal the study does not say that surveys have already been done or that a group of participants has been chosen. Before collecting any data clear steps need to be set up. These steps should cover how to pick participants why a certain number of people is needed how to get consent what language to use when talking to people, where the study will take place and what role agricultural experts will play.

Including experts would help check important things like planting thresholds how to prepare the land good soil management methods and whether the estimated costs and incomes are realistic. Having farmers take part would also show whether the system's interface is easy to use and if the input requirements make sense. It would also help see if the advice the system gives is practical, in farming situations.

Any results based on expert opinions or farmer experiences should only be shared once solid evidence and written feedback have been collected [2].

3.3 Experimental or study procedure

3.3.1 Step 1 — Plot selection and image preparation

At the beginning the user chooses the area where they want to grow crops. The user can do this by drawing a polygon on the map interface or by uploading a picture and drawing the boundary themselves. The system then checks the chosen polygon for correctness making sure the polygon does not cross itself. If good geospatial data

exists, the system calculates the land area automatically. The polygon stays separate from the RGB image that the system later uses for analysis. When a normal photo has no reference the user must type the real land size manually because pixel size alone cannot give real-world measurements. Any outside image that the user adds must follow the license rules for the analysis. Google Earth Engine can export images for chosen areas and scales [9]. The Copernicus Data Space Sentinel Hub API can also provide image processing and data access for satellite data [12]. Therefore, before adding an image source the user must check that the data is available and that the user has permission to use it. I find that drawing the polygon directly on the map makes it easier to see the area.

After the picture arrives the system decodes it. Corrects the orientation. Then the system checks that the picture size and file size are okay. The polygon coordinates defined by the user become pixel coordinates. Form a binary mask. Even though the model works on a RGB image only the pixels that lie inside the polygon count for land-class calculations. Pixels outside the polygon are ignored for the numbers. The system keeps information such, as the image source, the date it was taken the resolution, the model name and version the original class probabilities and any warnings. This helps the user trace what happened [4].

3.3.2 Step 2 — Land-cover segmentation and preliminary capacity

Per-pixel land-cover probabilities are generated by Model 1. In order to compute class percentages and estimated areas, the system aggregates valid pixels inside the chosen region. Low-confidence or unknown pixels should be communicated rather than quietly assigned to usable land, and the percentages should add up to around 100% within a specified rounding tolerance. After access and buffer deductions, confirmed agricultural pixels are used to calculate the ready area. Only after preparation, soil, water, and cost analysis is barren or open land deemed potentially convertible. [4]
Plant spacing, recorded row spacing, and a field-efficiency factor are used in an initial plant-position estimation. Positions can be expressed as follows: floor (usable area × field-efficiency factor ÷ (row spacing × plant spacing)). Crop/variety and source version must be connected to spacing settings. The method displays a range or tentative

estimate until the pertinent agronomic value is proven because general chilli/capsicum spacings might not be validated for the chosen Scotch Bonnet variety [29]. The final viable plant count and final suitability class cannot be issued by Model 1.

3.3.3 Step 3 — Two-photo preparation analysis

A device identifier, observation timestamp, sequence number, units, calibration version, and device status are all linked to each sensor payload. Sequence numbers and duplicate timestamps are handled impotently. The backend computes rolling summaries over a predetermined time period, saves raw and converted measurements, verifies reasonable physical constraints, and detects spikes or stuck sensors. Readings that are missing or outdated are identified as such and are not substituted with zero.

The appropriateness gate verifies that the necessary evidence—temperature history, water dependability, drainage, soil texture, pH, and salinity/EC when applicable—is present and up to date. The system responds "PROVISIONAL – NOT ASSIGNED," displays the missing components, and offers only safe interim measures if the necessary evidence is missing. The four existing IoT measurements cannot be used to infer soil texture, pH, or EC. In accordance with the FAO-style approach, each component is compared to versioned crop-use levels whenever the data is sufficient, and the most limiting verified factor determines the overall class [2].

3.3.5 Step 5 — Resource and feasible-count calculation

A budget, employee count, available working hours, owned equipment, water supply, and water reliability are all entered by the user. Although fuel, energy, maintenance, and consumables may still be included, the system does not include a purchase cost for equipment that already exists. Purchasing, renting, and outsourcing options are evaluated for missing items. Seasonal operating expenses and reusable capital expenditures are kept apart.

The minimum of the farmer-planned count and the capacities supported by actual land, soil, or a chosen cultivation method, as well as the budget, labor, and water, constitute the ultimate possible plant count. When a constraint fails, the system describes the limiting issue and suggests a supported course of action, including lowering the number of plants, renting equipment, extending the timetable, enhancing water

storage, or assessing an alternative bag-cultivation strategy. Bag cultivation cannot be regarded by the system as a modification to the initial soil suitability class.

3.3.6 Step 6 — Financial estimation and report generation

The financial engine integrates yield, survival, quality, and sale-price assumptions with preparatory costs, planting materials, equipment, irrigation, personnel, and operating inputs. It separates retail observations from farm-gate, contract, and wholesale pricing. Proxy values must be identified and given lesser confidence when Nai Miris pricing evidence is not available. Units, source, date, place, and evidence type are recorded together with costs, yields, and prices.

In order to assess profit, loss likelihood, budget-overrun probability, break-even output, and return on investment (ROI), the risk-aware stage samples recorded low/base/high ranges or probability distributions. When simulation repeatability is needed, the random seed and parameter versions should be recorded by the model. Income is not assured; outputs are ranges and probabilities. The chosen plot, evidence completeness, picture information, model versions, appropriateness status, plant-count limitations, assumptions, warnings, and sources are all documented in the report that may be downloaded. [5]

3.3.7 Model development and controlled comparisons

Keep a manifest with the dataset version, source picture, site or region, class mapping, and split before training. Prior to tiling or augmentation, divide by farm or regional acquisition group. Neighboring or nearly identical tiles from a single original image cannot be found in both training and test sets. While model selection, probability thresholds, and two-photo fusion rules are adjusted using validation data, the final test set is kept locked. [6]

The architecture, encoder initialization, input size, normalization, batch size, optimizer, learning-rate schedule, loss, class weighting, epoch limit, and checkpoint-selection metric must all be specified in the training record. A model filename is not used to rebuild these values. Interrupted training is supported by check pointing after each epoch; optimizer state and epoch must be included in resume records. To reveal

domain change, ratings from public datasets and distinct Sri Lankan sites are provided separately.

Establish multi-label annotation guidelines for the seven conditions for ground photos, and use a field reviewer to resolve disputes. Throughout all splits, maintain the opposite-side photo pairing. On the same held-out pairs, compare the mean probability, maximum probability, and noisy-OR. Since overlapping views are connected and may exaggerate perceived confidence, noisy-OR is merely a candidate fusion rule. Adjust criteria based on validation data and report the impact on retake rate and false negatives.

Create an expert-reviewed case set for decision logic that includes all necessary information, entire evidence, stale sensors, implausible units, inconsistent soil results, and expensive alternatives. Keep the system outcome and the anticipated result apart. Prior to field evaluation, recruitment and power or precision rationale must be accepted; no arbitrary accuracy target or finished sample size is claimed.

3.4 Measures and evaluation metrics

Each subsystem and the entire decision workflow should be evaluated independently. Prior to testing, metrics must be chosen and computed using independently verified observations or reference labels. [3]

Table 3-2 Subsystem evaluation measures

Subsystem	Evaluation measure	Purpose / interpretation
Land-cover segmentation	Per-class precision, recall, F1-score, intersection over union (IoU), confusion matrix	Shows which land-cover classes are confused; class-specific reporting is needed because errors have different consequences.
Area and capacity estimation	Absolute and relative area error; planting-capacity error against manually verified boundaries and spacing	Measures whether segmentation and boundary errors materially change usable area and estimated plant positions.
Ground-photo model	Per-class precision/recall/F1; quality-gate failure rate; retake rate	Evaluates visible-condition detection and whether poor photographs are handled consistently.
Farmer confirmation	Agreement across users; frequency and magnitude of edits; completion time	Assesses interpretability and consistency of affected-fraction confirmation.

Subsystem	Evaluation measure	Purpose / interpretation
IoT pipeline	Missing/stale reading detection; calibration error where a reference instrument exists; duplicate rejection	Evaluates data integrity and reliability of sensor summaries.
Suitability gate	Agreement with expert-reviewed cases; rate of unsupported final classifications; missing-evidence detection	Checks whether the system correctly withholds a final class when evidence is incomplete.
Financial engine	Break-even calculation checks; forecast error when actual farm records become available; calibration of loss probabilities	Assesses arithmetic correctness and, later, whether estimated ranges reflect observed outcomes.
Integrated workflow	Constraint-violation rate; report completeness; task completion; user comprehension and perceived usefulness	Evaluates whether recommendations respect constraints and whether users understand limitations.

A single total score should not be used to summarize accuracy. For instance, a building's useable area may be overstated if it is incorrectly classified as agricultural, and preparation costs may be primarily impacted if two vegetation types are confused. Therefore, in addition to traditional model measures, error severity and downstream impact should be taken into account. Only when there are enough local farm records available should financial forecasts be evaluated; in the interim, the report can confirm calculation accuracy and sensitivity to assumptions but cannot guarantee predictive accuracy. [3]

3.4.1 Metric definitions and interpretation

Precision = TP/(TP + FP), recall = TP/(TP + FN), F1 = 2TP/(2TP + FP + FN), and IoU = TP/(TP + FP + FN) for each class. The unweighted mean over the designated evaluated classes is the mean IoU. How absent classes and ignored pixels are handled must be specified in the evaluation report. Rare classes still have an impact on macro averages, although aggregate pixel precision can conceal their mistakes.

The mean absolute difference in m^2 between the projected and reference class areas is known as area MAE. Only reference areas that are not zero are used to report relative area inaccuracy. Report the whole expert-versus-system confusion matrix and the number of final classes that were provided in spite of insufficient evidence for

appropriateness; zero unsupported final classifications is the expected invariant. This is not a quantifiable outcome, but a prerequisite.

The percentage of observed outcomes that fall within a forecast P10–P90 interval is known as empirical coverage; a properly calibrated interval should cover about 80% of comparable events. To avoid trivially broad predictions looking useful, report interval width along with coverage. Use chronological holdout for financial back-tests and farm-level resampling for uncertainty intervals when many records originate from a single farm.

3.5 Data analysis

To minimize leakage from nearby or nearly identical images, held-out test data that are isolated from training and validation data at a suitable geographic or farm level should be used for quantitative model evaluation. For segmentation tasks, per-class confusion matrices as well as precision, recall, F1-score, and IoU should be reported. Land-cover annotations and manually confirmed reference boundaries should be used to determine area-estimation errors. When sample size allows, repeated runs or confidence intervals should be used. [4]

When it comes to IoT validation, sensor readings should be checked with appropriate reference tools or, if possible, laboratory or field observations. The percentage of stale or missing readings found, duplicate events rejected, and sensor defects reported should all be considered data-quality outcomes. In order to confirm that a final class is withheld when necessary evidence is missing, suitability-gate testing should incorporate both complete and purposefully incomplete evidence examples.

Deterministic unit tests for formulas, units, and constraints should be the first step in financial evaluation. Sensitivity analysis should determine which assumptions have the biggest impact on ROI and break-even. Forecast errors and the calibration of expected loss probabilities can be assessed if enough observed farm data later becomes available. Until then, simulated results should not be used as evidence of predicted revenue, but rather as scenario analysis based on given assumptions.

Thematic analysis of qualitative data from farmer and expert sessions should focus on usability, warning clarity, trust, perceived utility, language accessibility, and whether users comprehend that the tool enhances rather than substitutes agronomic judgement.

Only once the sessions have actually taken place can the number of participants, recruitment strategy, coding process, and representative findings be provided.

3.6 Ethics, privacy, security and responsible AI

Plot boundaries, location, land photos, farm resources, financial data, and sensor observations can all be processed by the system. Even if these data are not personally identifiable, they may nonetheless be financially sensitive. Users should be made aware of what is gathered, why it is necessary, how long it is kept, who can access it, and whether it will be used to create models. When necessary, consent should be acquired, and the system should only gather information essential to support decisions [3].

Authenticated access, role-based authorisation, secure transportation, protected storage, regulated report sharing, input validation, backups, and audit records for significant modifications are all examples of security controls. Verify the file type, dimensions, size, and faulty content of uploaded photos and geospatial files. In addition to authenticating devices, device ingestion should guard against duplicate or replay communications. Prior to field implementation, policies for data retention and destruction should be established [3].

Transparency on model limitations, source data, confidence, and downstream effects is necessary for responsible AI. Validity and dependability, safety, security and resilience, accountability and transparency, explain ability, privacy enhancement, and management of harmful bias are among the attributes listed in the NIST AI Risk Management Framework [3]. Visible warnings, source traceability, manual review for uncertain or high-impact scenarios, and the capacity to differentiate model output from measurement or assumption are all examples of how these concepts apply to NAIMIRIS. It is not appropriate to automate recommendations for chemical application, hazardous debris clearance, or tree removal based solely on image class. Crop-specific thresholds and practical recommendations must be reviewed by agricultural specialists.

Users with varying degrees of digital literacy should be able to access the system. Barriers can be lowered via clear language, comprehensible units, support for local languages, low bandwidth operation when practical, and definitions of new words.

When money, water, or labor inputs change, a user should be able to comprehend why a recommendation changes and modify farmer-entered data [3].

3.7 Reproducibility and research artefacts

Each analysis should preserve the input identifiers, imagery source and capture date, polygon geometry version, preprocessing parameters, model name and version, class mapping, thresholds, calibration version, source values, cost and market-data timestamps, assumption versions, and decision-rule version in order to make the research reproducible. The random seed and distributions should be documented when using simulation. The system should preserve the justification for a manual override and differentiate between initial user entries and subsequent corrections.

A data dictionary, model cards or comparable model documentation, dataset license records, annotation instructions, API/input-output contracts, decision-rule documentation, test cases, evaluation scripts, risk register, and sample anonymized reports are examples of research artefacts. Farm data that is personally or commercially sensitive shouldn't be stored in a public repository. It is important to document and adhere to any license or attribution obligations imposed by source datasets. [4]

Only when actual implementation and evaluation artefacts are available should they be included in the final thesis. It is necessary to designate as a design illustration rather than actual evidence any suggested architecture diagram, simulated scenario, or sample JSON payload. Reproducibility encompasses not just successful cases but also failures and missing evidence.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

4.1.1 Functional requirements

Table 4-0-1 Functional requirements

ID	Requirement	Expected behavior
FR-01	Plot input	Allow map-assisted polygon selection or image upload and boundary drawing; validate geometry and preserve ROI separately.
FR-02	Land-cover analysis	Classify overhead imagery, calculate class proportions/areas within the ROI and return metadata, confidence and warnings.
FR-03	Preliminary capacity	Estimate a planting-capacity range from approved usable area, documented spacing and field-efficiency assumptions.
FR-04	Ground-photo analysis	Accept exactly two guided opposite-side photos, evaluate quality and classify visible preparation conditions.
FR-05	Farmer confirmation	Request confirmation of affected-area bands and allow a farmer to review or override the proposed plant count.
FR-06	IoT ingestion	Store raw and calibrated readings with timestamps, units, device and calibration metadata; detect stale, duplicate or implausible data.
FR-07	Suitability evaluation	Apply a limiting-factor framework and return a provisional status when required evidence is missing or stale.
FR-08	Resource inventory	Capture budget, workers, available hours, owned equipment, water source/reliability and other relevant resources.
FR-09	Feasible plant count	Constrain the plan by physical land, soil/method, budget, labour and water capacity.
FR-10	Financial estimation	Estimate cost, break-even, ROI and risk ranges from versioned evidence and assumptions; never guarantee income.
FR-11	Alternative plans	Compare open-field, bag, shade-net or greenhouse/protected options where sufficient evidence and cost data exist.
FR-12	Traceable report	Generate a report with evidence completeness, warnings, sources, dates, model versions, assumptions and confidence.

4.1.2 Non-functional requirements

Table 4-2 Non-fuction requirement

ID	Requirement	Design implication
NFR-01	Transparency	Separate model output, IoT measurement, farmer input, sourced value, proxy and assumption.
NFR-02	Reliability	Validate input files, polygons, sensor payloads, units and decision-rule preconditions.
NFR-03	Security and privacy	Use authentication, authorization, protected transport/storage, controlled sharing and retention rules.
NFR-04	Usability	Use guided photo capture, clear units, concise explanations, accessible language and correction paths.
NFR-05	Reproducibility	Version models, thresholds, costs, source data, calibration and simulation parameters.
NFR-06	Robustness	Handle missing, stale, contradictory and low-confidence evidence without fabricating values.
NFR-07	Maintainability	Keep image models, evidence ingestion, suitability rules and financial calculations modular.
NFR-08	Responsible recommendations	Escalate uncertain, hazardous or agronomically consequential decisions to manual/expert review.

The backend definition and proposal objectives serve as the basis for the functional requirements. They do not assert that every feature has previously been implemented or passed testing; rather, they specify the desired behavior of the system.

4.2 Component architecture within the complete solution

The structure of NAIMIRIS is a phased pipeline for decision support. A traceable pre-cultivation plan is returned by each component after processing inputs from the user and outside evidence sources using modular services. The cultivation-guidance component of the group may eventually use its outputs, but the implementation of the other group members' components is not specified in this report.

Table 4-3 Service responsibilities

Layer / module	Inputs	Processing responsibilities	Outputs
1. User input and plot selection	Location, polygon or uploaded image, land size, farmer details	Validate inputs; calculate/confirm area; retain geometry and source metadata	Validated plot and user-input record
2. Imagery acquisition and preprocessing	Plot geometry, permitted image source or uploaded image	Acquire permitted image; orient, crop, resize/tile, rasterize ROI mask	Model-ready RGB crop, ROI mask, image metadata
3. Model 1 land-cover service	RGB crop and ROI mask	Semantic segmentation, confidence thresholding, class aggregation	Class probabilities, class areas, unknown percentage, preliminary capacity
4. Model 2 preparation service	Exactly two guided ground photographs	Quality scoring, condition classification, cross-photo combination	Condition probabilities, severity and retake/manual-review flags
5. IoT and soil-evidence service	Sensor payloads, lab reports, soil maps, drainage/water evidence	Authenticate, calibrate, quality-check and summarize time-series evidence	Validated observations and evidence-completeness state
6. Suitability decision engine	Land evidence, crop references, soil/water evidence	Evaluate limiting factors; separate current and potential suitability	Suitability class or provisional status, limiting factors, missing evidence
7. Resource and cost engine	Budget, labour, equipment, preparation tasks and unit costs	Calculate buy/rent/outsource options, task hours, capex and opex	Preparation plan and resource capacity
8. Feasibility and financial engine	Plant count, water, labor, costs, yields and price ranges	Apply minimum-capacity constraint; break-even and risk simulation	Feasible count, alternatives, profit/loss and ROI ranges
9. Report and audit layer	All module outputs and provenance metadata	Assemble source trace, assumptions, warnings, versions and results	Downloadable decision report and audit record

Figure 4-3 Service responsibilities

The architecture keeps the financial and suitability engines apart from the image-model services. This stops budgetary restrictions or missing soil evidence from being circumvented by a high-confidence visual classification. Additionally, it makes it possible to swap out or retrain a model without subtly altering the downstream decision rules. In order to track a recommendation back to the assumptions and inputs that led to it, the report layer maintains provenance.

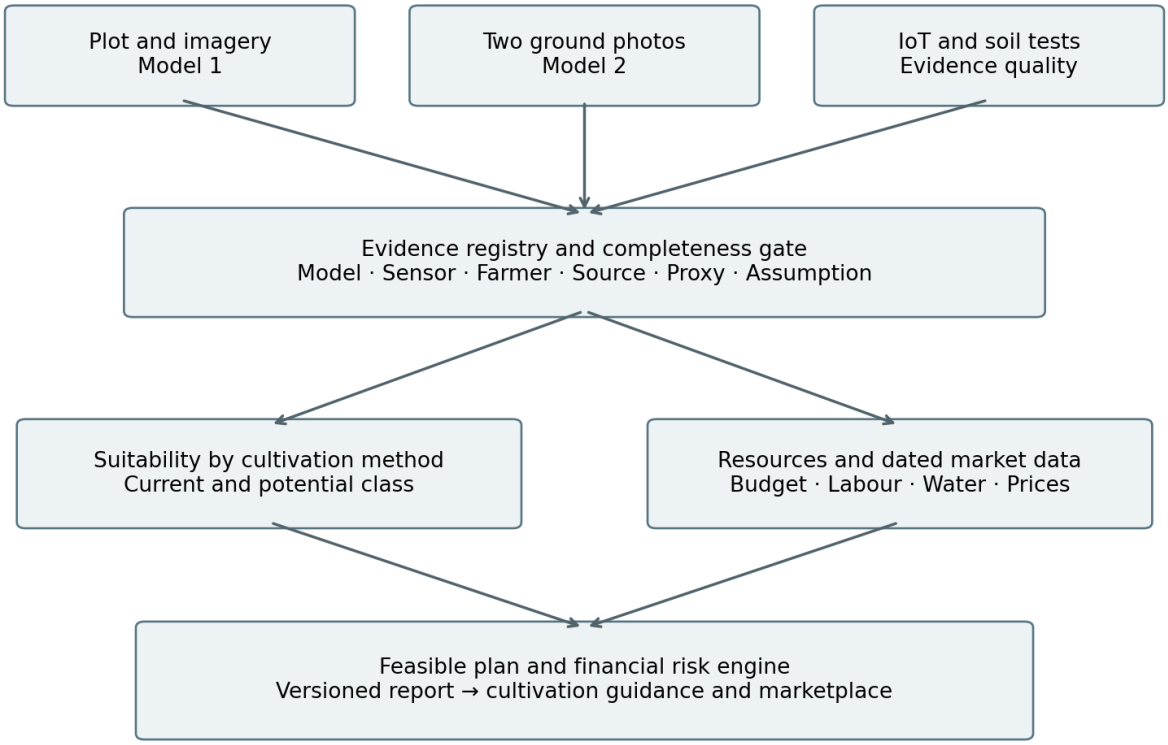


Figure 4-1 Component architecture

Figure 4.1 Component architecture and shared evidence registry

Service duties within each component are depicted in Figure 4.1. The approved technique, planting count, preparation tasks, and evidentiary limitations are sent to the guidance module. Instead of guaranteeing a future sale price, marketplace or price-intelligence services offer channel-labeled observations. An authorized plan snapshot may be consumed by other group modules, but its assumptions and supporting data are not quietly overwritten.

4.3 Design alternatives and decisions

Table 4-4 Design alternatives

Design question	Alternative considered	Selected design decision	Reason
Plot acquisition	Uploaded image only; map-only; hybrid	Hybrid map-assisted polygon selection with image-upload fallback	Supports user choice and fallback when imagery is unavailable or unsuitable; normal images require land size if not georeferenced.
ROI handling	Irregular transparent crop; rectangular crop plus mask	Rectangular RGB crop plus separate binary ROI mask	Keeps model input conventional while ensuring only pixels inside the selected boundary contribute to area calculations.
Ground images	One image; unrestricted uploads; guided pair	Exactly two guided opposite-side photos	Improves coverage consistency while acknowledging that two views do not provide survey-grade area measurements.
Soil evidence	Infer soil type from current IoT; require independent evidence	Require independent texture/pH/EC evidence where relevant	The four current sensor variables cannot establish these properties.
Suitability	Always return best-guess class; gate missing evidence	Return PROVISIONAL – NOT ASSIGNED when required evidence is missing	Avoids unsupported agronomic conclusions and identifies next evidence needed.
Plant count	Use land capacity alone; use a multi-constraint limit	Minimum of physical, soil/method, budget, labour and water capacities	A physically possible plan may not be operationally feasible.
Financial output	Single point estimate; low/base/high and probability ranges	Ranges and risk probabilities with traceable assumptions	Reflects uncertainty in yield, price, cost and labour; avoids implying guaranteed income.
Evidence provenance	Store final values only; preserve source categories and versions	Preserve model, sensor, farmer, sourced, proxy and assumption categories	Allows audit, correction, confidence interpretation and reproducibility.

Transparency and safe failure are given precedence over a seemingly straightforward result in these choices. The primary drawback is that a gated system could request additional information from the user and provide a tentative outcome rather than a final recommendation. This is deliberate: rather than hiding uncertainty, the system should disclose it when it could significantly alter a farmer's investment decision due to missing evidence. [3]

4.4 Detailed design

4.4.1 Model 1: overhead land analyzer

A rectangular RGB image and a certified polygon are sent to the land analyser. The canonical NAIMIRIS classes—agriculture, barren/open, building, road/access, water, forest/trees, developed/other, and unknown—are mapped into the source-dataset classes. Because multiple taxonomies are used in the source datasets, class mapping needs to be versioned. The model applies a low-confidence threshold, keeps the per-pixel probabilities, and labels pixels outside the ROI as ignored. [6]

Each class percentage for a uniformly sampled, fully observed polygon is calculated by dividing the number of inside-polygon pixels by the total number of inside-polygon pixels. Instead of being redistributed among known classes, unknown and missing coverage stays explicit. Variable pixel areas are handled by the area-weighted equation that follows. Barren land retains prospective until preparation and agronomic viability are assessed; ready capacity employs confirmed usable area after access and buffer deductions. [4]

An analysis identifier, input technique, chosen area, land-cover results, ready and potential capacity ranges, unknown percentage, confidence, imaging information, model metadata, and warnings are all included in a conceptual response record. The answer shouldn't suggest that Model 1 by itself determines the ultimate plant count or FAO class.

Table 4-5 Design alternatives

Canonical class	Dataset mapping	Decision treatment
Agriculture	LoveDA agriculture; OEM agriculture	Candidate area subject to verification
Barren or open	LoveDA barren; OEM bareland	Potential only after preparation and suitability
Building and road	Corresponding source classes	Exclude from planting
Water	Corresponding source class	Exclude and investigate drainage
Forest or trees	LoveDA forest; OEM tree	Review and protect; no automatic clearing
Developed or other	LoveDA background; OEM developed space	Review or exclude
Rangeland	OEM rangeland	Keep distinct during mapping; no automatic barren conversion

Canonical class	Dataset mapping	Decision treatment
Unknown or no data	Confidence failure or missing imagery	Report separately; never count as ready

Unmapped pixels cannot be redistributed among recognised classes by area aggregation. Let w_i be the geographic area that pixel i in the polygon represents, and let A be the polygon's area. P_c = 100 A_c/A; class area A_c = Σ w_i I(label_i = c). To ensure that the totals accurately reflect the entire polygon, explicitly include unknown and no-data coverage. This reduces to a pixel-count ratio for a uniform equal-area raster; area weighting prevents edge and latitude bias for geographic or resampled grids.

The ROI mask and crop picture employ the same transformation. Nearest-neighbor resampling is used for categorical labels and masks; picture interpolation comes after training preprocessing. Edge pixels are handled by either documented rasterisation tolerance or fractional boundary coverage. User-supplied total area only supports an approximate scaling assumption for a non-georeferenced overhead image. Reliable overhead class areas cannot be produced from an oblique snapshot just because a land-size number was entered.

Ready capacity is calculated as floor (A_ready × e / (s_row × s_plant)), where spacing is measured in meters and e is a specified layout efficiency. Access zones that have been specifically eliminated cannot be subtracted using e. The configured square-perch conversion needs to be visible and versioned, but the interface can display perches while keeping m² internally. Until layout validation is finished, the area-based estimate may be an upper bound due to plot form, setbacks, and access paths.

4.4.2 Model 2: two-photo preparation analyser

Two guided photos are accepted by the ground analyser. Blur, exposure, duplicate viewpoint, and useable scene fraction are all scored for each picture. If a picture doesn't work, just that picture is asked for once more. The model calculates the likelihood that each visible situation will exist. Condition presence can be coupled using a noisy-OR combination, and severity can be combined using quality-weighted scores. The condition is marked for farmer confirmation if the two photos show a significant discrepancy.

Instead of producing a precise impacted region, the model produces condition probability and severity. Using candidate land area and an explicit fraction mapping, the farmer verifies an impacted fraction band, which is then transformed into a documented area range. Task hours are calculated using a calibrated hours-per-square-meter coefficient, and preparation costs are calculated by multiplying the impacted area by a specified unit-cost range. Trees are not automatically cleared and must be reviewed by hand. Handling potentially dangerous debris calls for expertise.

The candidate fusion score for class c is equal to $p_c = 1 - (1 - p_A,c)(1 - p_B,c)$. Only if there is a calibrated severity output and $q_A + q_B > 0$ can a quality-weighted severity be computed. Instead of establishing a verified severity estimator, the source files classify conditions; in other cases, ask for farmer-confirmed severity rather than creating it. In order to avoid charging for the same clearing activity twice, overlapping grass and brush regions must share task identifiers.

4.4.3 IoT data contract and quality controls

A device identifier, timestamp, sequence number, readings, units, calibration version, battery status where available, and authentication details are all included in each IoT payload. The service rejects duplicate or incorrect events and maintains both converted and raw values. Rolling summaries, spike detection, stuck-sensor detection, and physical plausibility tests are used across predetermined windows. If a sensor reading is older than the configured age or if the observation period is too short, it is considered stale.

The evidence service keeps track of whether necessary environmental and soil inputs are present or not. It cannot replace missing data on pH, EC, soil texture, drainage, or water reliability with default values. Until the necessary evidence is sufficient, the system returns a provisional suitability status even if it may display interim observations.

4.4.4 Suitability decision logic

Each confirmed land feature is compared to a versioned requirement for the chosen crop and cultivation technique by the appropriateness engine. The FAO-style definitions given in Section 2.2.1 [2] apply to the classes S1, S2, S3, N1, and N2. The

most severe verified limiting factor determines the overall class, and the system logs the factor and supporting data. Potential suitability following recorded improvements is kept apart from current appropriateness.

The engine returns "PROVISIONAL – NOT ASSIGNED" and lists missing evidence, noted limitations, and safe interim actions where any necessary evidence is absent or outdated. Image confidence alone does not yield a final class. Options for improving drainage or adding organic matter to the soil are considered expenditures and, where necessary, require agronomic approval. Protected cultivation, also known as bag cultivation, is considered as a distinct approach with its own budget, inputs, and capacity. [2]

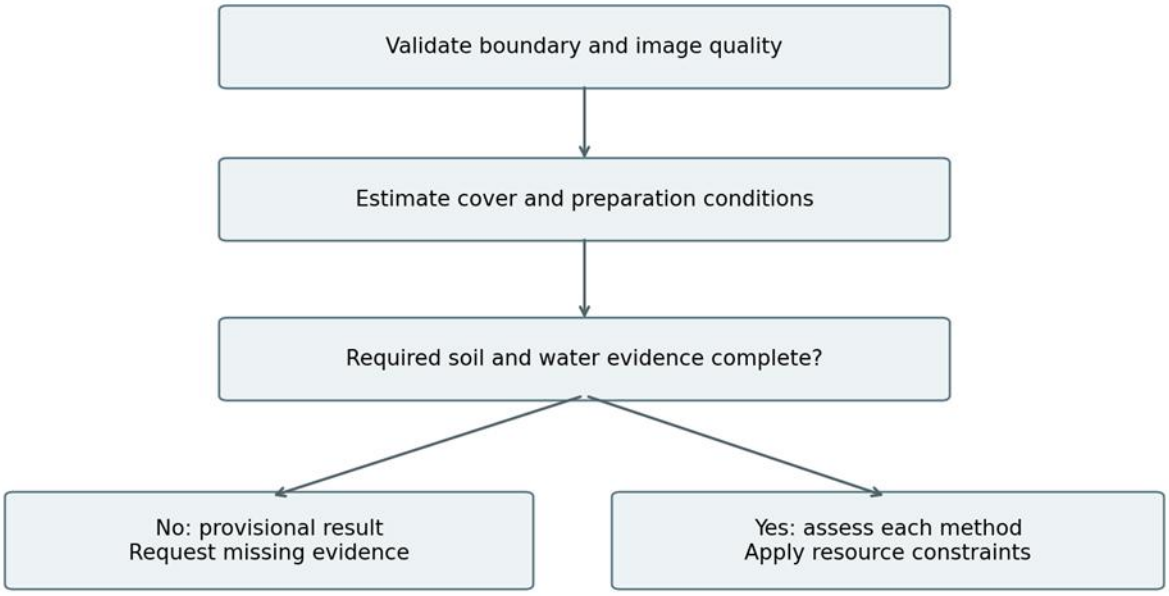


Figure 4-2 Decision logic

Figure 4.2 Evidence completeness and suitability decision gates

Insufficient proof and inappropriate land are separated in Figure 4.2. A pH result that is absent does not qualify as N1 or N2. Each cultivation method's necessary elements, acceptable age and source quality, and agronomist-approved restrictions are all outlined in the rulebook. N2 does not refer to a permanent ban on all potential uses, but rather to the established land use and evaluation assumptions [2].

4.4.5 Feasible plant count and resource engine

After estimating physical capacity, the backend verifies capacity under the chosen soil or cultivation technique, budget, manpower, and water. Min (farmer-planned count, approved physical capacity, soil/method capacity, budget capacity, labour capacity, and water capacity) is the ultimate viable count. There is a documented computation and failure explanation for each capacity. For instance, water capacity is based on dependable usable water in relation to demand under the chosen method; labour capacity is based on available worker-hours compared with fixed and per-plant task hours; and budget capacity is the greatest number of plants whose cash requirement stays within the available budget and reserve.

Owned, buy, rent, and outsourced alternatives are differentiated in the resource catalogue. Equipment that is already owned is not subject to purchase expenses; but, recurrent fuel, energy, maintenance, or consumable expenditures may still be incurred. Seasonal operating expenses and capital expenditures are kept apart. Instead than presenting equipment ownership as a binary yes/no that completely determines feasibility, alternatives are assessed using total cost and practical limits.

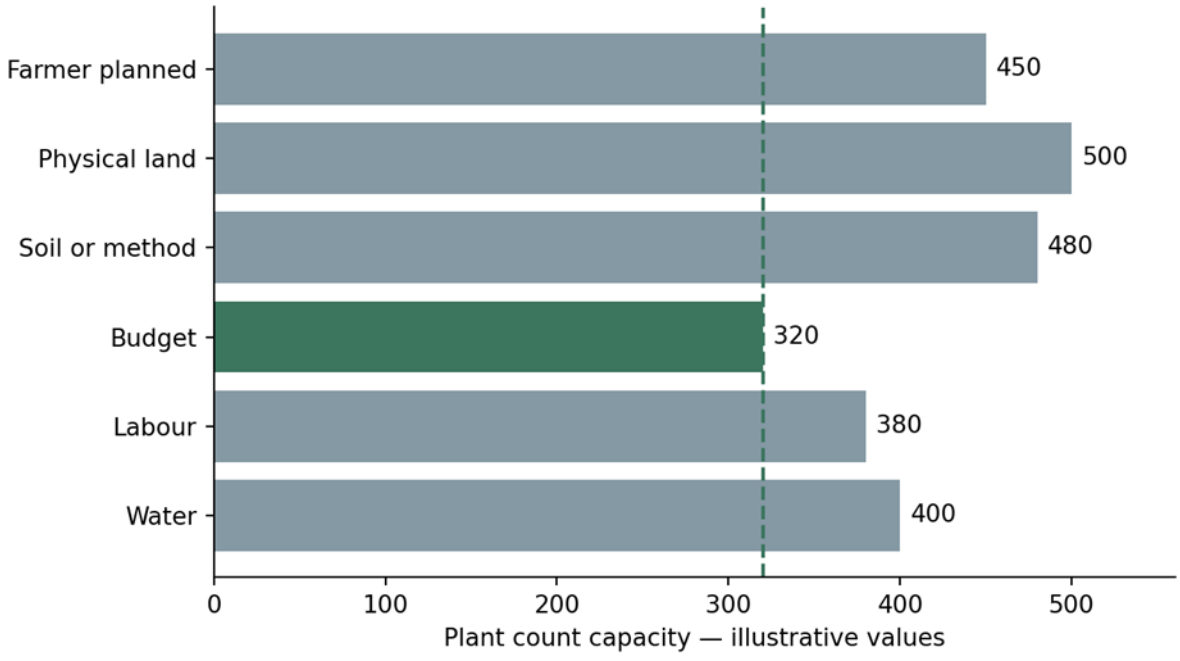


Figure 4-3 Plant count capacity

Figure 4.3 Illustrative plant-count constraint calculation

The minimum-capacity rule is illustrated in Figure 4.3 using only created design inputs: min(450, 500, 480, 320, 380, 400) equals 320 plants, with budget binding. These measurements are not taken on a farm. A failing appropriateness gate cannot be circumvented by overriding the anticipated count, but the user can compare resource changes.

Capacity depends on both time and method. Compute fixed preparation expenditures, incremental equipment purchases, seasonal task hours, and water requirements per stage for each candidate count. Only in cases when each constraint specifies a feasible interval from zero to its upper bound is the minimum of scalar capacity valid. Count the number of candidates for non-monotonic alternatives or fixed-cost discontinuities, and keep only those plans that pass every test. If transplanting or peak-harvest weeks surpass available capacity, total seasonal labor and water are insufficient.

4.4.6 Financial model and uncertainty

The financial engine determines the projected salable yield range, revenue range, break-even point, ROI range, and itemized preparation and cultivation costs. It captures the date, region, unit, and source for every price and makes use of the proper farmer revenue route. A proxy is clearly identified and should lower trust when local Nai Miris prices are not available. Retail and farm-gate pricing cannot be combined in the system as if they were equal.

Uncertain variables are sampled by the risk engine from low/base/high ranges or established distributions. ROI percentiles, median profit, probability of loss, and probability of budget overrun are examples of outputs. Each report must display the main drivers and provide the user with the ability to review or modify assumptions because the results are dependent on the assumptions made. Simulation does not promise financial gain; rather, it is a tool for decision-making. Before making significant predictive claims, the model should be calibrated against local yield, survival, quality, labour, and sales records over several seasons. [5]

4.4.6.1 Cash flow and economic accounting

One specified agricultural cycle or season is the accounting horizon. Let D represent owned-asset usage and capital consumption, K represent new capital purchases, O

represent seasonal cash operating expenditures, and L represent imputed family labour. K + O is the first-season cash outflow. O + L + D = seasonal economic cost. When its assigned consumption is already included in D, the entire purchase K is not added to the seasonal economic cost once more. Although purchasing new equipment is avoided, running and use charges are still justified.

Planted count × survival fraction × yield per surviving plant × quality acceptance fraction equals salable output Q. Survival cannot be applied twice once yield per planted plant has already been recorded. Using the corresponding harvest month and farmer selling channel, monthly revenue $R = \Sigma(Q_m \times P_m)$. Before wholesale prices reflect net farmer receipts, specific deductions for transportation, grading, commissions, and losses must be made.

Benefit-cost ratio = R/C_e; break-even price = C_e/Q; economic profit = $R - C_e$; economic ROI (%) = $100(R - C_e)/C_e$. In this case, Q is salable kg and C_e is seasonal economic cost. The equivalent ratio is undefined, not 0, when there is no cost or output. In a fixed-cost simplification, break-even yield = C_e/P is valid; if marketing and harvesting costs change with Q, solve $R(Q) - C_e(Q) = 0$.

Cash viability is monitored over time. Required financing = $\max(0, -\min_t CF_t)$ with cumulative net cash flow CF_t. A seasonal surplus does not demonstrate the farmer's ability to cover preharvest preparatory expenses. Until it is used, a visible contingency reserve stays a budget buffer rather than an actual economic expense. A reserve and the same sampled cost-overrun allowance should not be duplicate counted.

4.4.6.2 Price evidence and risk simulation

Commodity, variety or grade, district, buyer or market, date, channel, amount, and LKR/kg conversion are all retained in price observations. When amounts are available and comparable, aggregate transaction prices by quantity; otherwise, use a verified median with sample count and dispersion. Missing months are still missing. A single published farm tale, store listings, and supplier quotes are insufficient to create a continuous Nai Miris farm-gate series.

10,000 draws with a saved seed are used in the specified Monte Carlo setup. Sample monthly price values from distributions supported by evidence, non-negative yield and costs, and bounded survival and grade fractions. Keep track of dependent assumptions,

particularly the effects of typical weather on labour, quality, and yield. When feasible, use empirical paired observations; in cases where the evidence is poor, stress-test probable relationships. The distribution choices here are project design options; the FAO discussion supports simulation as a risk-analysis technique [5].

Recalculate harvest, cash timing, economic cost, profit, and feasibility for every draw. P10, P50, P90, the likelihood of a profit, the likelihood of a loss, and the likelihood of remaining under budget should all be summarised. Prior to adjusting precision, check stability across seeds or additional draws. Although it cannot prove causal effects or validate the input distributions, Spearman rank sensitivity can find correlations in the simulation [5]

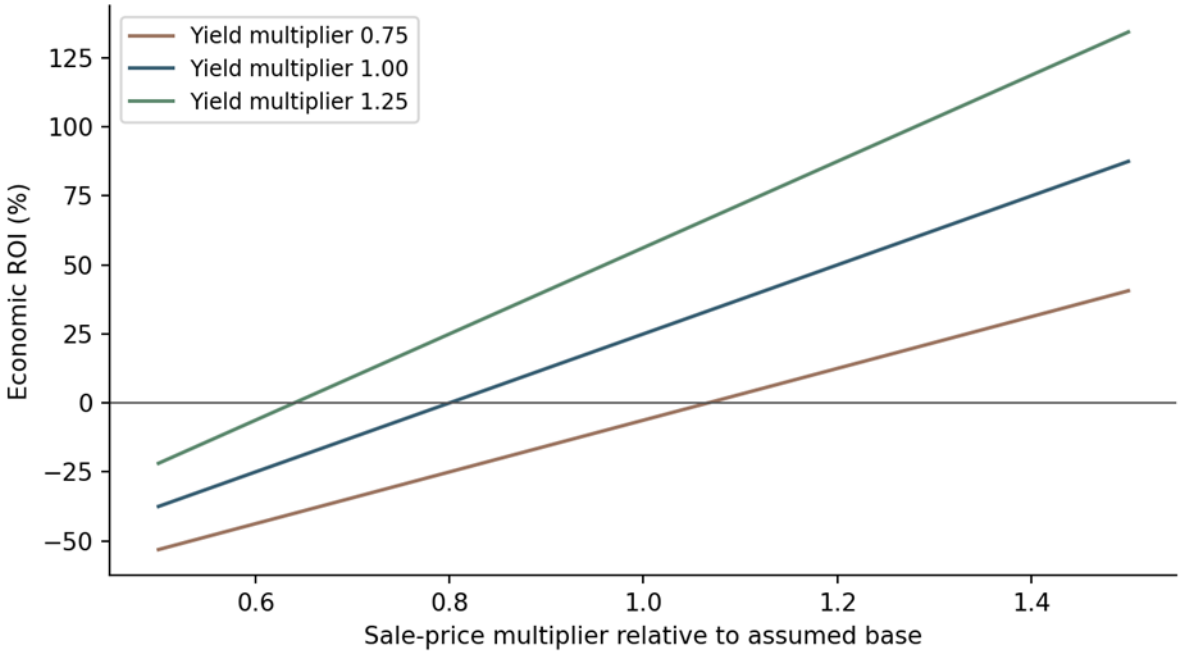


Figure 4-4 Sales price

Figure 4.4 Illustrative sensitivity of economic ROI to price and yield

An algebraic example is shown in Figure 4.4, where base income is assumed to be 1.25 and fixed economic cost is normalised to 1. Consequently, ROI = 100 (1.25 × yield multiplier × price multiplier − 1). A 25% yield reduction at the base price results in −6.25%, while the base point yields 25%. This demonstrates why one optimistic estimate may change under reasonable circumstances. There is no representation of the price, yield, profit, or accuracy result.

4.4.7 Report structure and traceability

The chosen plot and area, land-cover summary, model versions, ground-condition findings, user-confirmed affected fractions, IoT summaries, missing evidence, suitability status, planned and feasible plant counts, limiting constraints, suggested cultivation alternatives, resource and preparation requirements, itemised cost ranges, break-even and financial-risk ranges, source records, assumptions, warnings, and confidence should all be included in the generated report. Measured values should be distinguished from user input and model projections. Additionally, the report should indicate when a professional review is advised [3].

4.4.8 Error and exception handling

Table 4-6 Exception handling

Condition	System response
Invalid or self-intersecting polygon	Reject submission, explain the geometry problem and allow the user to edit the boundary.
Image is too old, cloudy, low-resolution or unlicensed for analysis	Warn or reject the image source; request permitted alternative imagery or an upload.
Low-confidence land-cover pixels	Retain as unknown, show the proportion and request review if it materially affects capacity.
One ground photo fails quality check	Request a retake of that photo only; do not silently add a third input.
Ground photos disagree materially	Flag the condition for farmer confirmation or manual review.
IoT reading is duplicate, stale or implausible	Reject or mark the reading, retain diagnostic metadata and do not replace it with zero.
Required soil or water evidence is missing	Return PROVISIONAL – NOT ASSIGNED and list the evidence needed.
Budget, labour or water constraint fails	Reduce the feasible count or compare documented alternatives; explain the binding constraint.
No reliable Nai Miris farm-gate price is available	Use no fabricated price; optionally show a labelled proxy with reduced confidence.
Hazardous debris or tree removal is indicated	Require manual/specialist assessment; do not automate removal instructions.

4.4.9 Service contracts and asynchronous execution

The backend specification suggests a geographical database and FastAPI-style APIs; these are design decisions rather than proof of a deployed technological stack. An authorized actor, the schema version, unit-bearing values, and pertinent evidence IDs are included in every request. A job identifier for polling is produced by laborious imaging, inference, and simulation. Impotency keys are used in retried procedures to prevent duplicating charges or records.

Table 4-7 Principal API contracts

Endpoint	Input and response	Validation
POST /v1/land-analyses/map	GeoJSON → analysis job	Geometry, ownership, licensed image source
POST /v1/land-analyses/upload	Image and mask coordinates → job	Decoded type, limits, georeferenced or area
POST /v1/ground-analyses	Photo A and Photo B → findings	Exactly two distinct quality-approved images
POST /v1/iot/readings	Signed observations → receipt	Device, sequence, calibration and units
POST /v1/suitability/evaluate	Evidence snapshot → class or provisional	Required evidence and versioned rules
PUT /v1/plans/{id}/plant-count	Integer choice → saved plan revision	Non-negative count, capacity explanation
POST /v1/plans/{id}/budget	Method and inventory → budget	No duplicate assets; dated cost sources
POST /v1/plans/{id}/simulate	Distributions and seed → job	Valid bounds; iteration and resource cap
GET /v1/plans/{id}/report	Plan revision → report snapshot	Owner access and immutable version

Jobs move through the following stages: VALIDATING, FETCHING_IMAGERY, PREPARING, RUNNING_MODEL, and CALCULATING; a recoverable workflow is paused by WAITING_FOR_FARMER; the job revision's explicit terminal outcomes are COMPLETED and FAILED. Identify unauthorized access from a missing resource, provide an actionable validation answer for an invalid polygon or incorrect photo count, and save finished upstream evidence in the event that a later step fails.

4.4.10 Data persistence and version boundaries

Table 4.8 summarizes core entities and reproducibility fields.

Table 4.8 Core entities and reproducibility fields

Entity group	Key records	Version boundary
Farm and selection	Owner, polygon, area, time zone	Boundary changes create a selection revision
Imagery and model run	Source, capture date, resolution, checksum, preprocessing	Each input/model combination is immutable
Ground and sensor evidence	Photo pair, condition probabilities, device, observation time	Raw data retained with calibration and quality flags
Suitability assessment	Method, factor classes, missing evidence, rule version	References exact evidence snapshot
Resources and observations	Inventory; prices; costs; source, date, channel	Dated observations are never overwritten
Plan and budget	Count, method, quantities, cash and economic lines	Farmer changes create a new plan revision
Simulation and report	Seed, distributions, results, report checksum	Report points to fixed versions of every input

A farm has numerous land selections and plans; a selection includes model runs and imaging assets; an accepted plan includes a budget modification, an optional simulation run, and one evidence snapshot. The suggested spatial store is PostgreSQL with PostGIS, while photos and generated reports are stored in protected object storage. Restricted object identifiers and checksums are used in database references. Financial information and identifiable coordinates are left out of public research exports.

4.4.11 Interface sequence and explanatory outputs

Plot selection, land summary, confirmation of plant count, two-photo capture, sensor and soil evidence, resources, technique comparison, and final report are the steps the farmer goes through. Each phase identifies necessary versus optional fields and maintains progress. Display workers and working hours separately, perches and m^2 together where applicable, and a date and channel next to each market value. Instead of a deceptive green appropriateness badge, an incomplete evaluation shows the thing that is missing and how to get it.

Using the same planned horizon and evidence snapshot, compare cultivation techniques. Current suitability, upgrades, first-season funding requirements, labour and water peaks, economic costs, risk, and unresolved evidence are all displayed on the method card. The current files do not allow an experimentally tested variety-ranking algorithm, therefore crop variety recommendations are still dependent on a supplied variety catalogue and locally evaluated criteria.

4.5 Standards and professional practice

Consistent data, software, and research techniques should be followed in the design. This report uses IEEE-style numbered citations, and each in-text citation is connected to a matching reference item. Explicit units and uniform timestamp formats should be used in all data fields. When available, coordinate reference data should be preserved in geospatial inputs, and the implementation should record the method used to determine geographic area. The licenses, attribution requirements, and terms of image suppliers and public datasets must be followed. The supported picture export and processing methods are described in the documentation for Google Earth Engine and Copernicus Sentinel Hub, although this does not eliminate the requirement to confirm the relevant keywords for a particular purpose [12].

Dataset provenance, label definitions, train/validation/test separation, versioned preprocessing, class-specific metrics, and limitation documentation should all be included in model-development procedures. Unless appropriate local experiments substantiate the assertion, a model trained with LoveDA or OpenEarthMap should not be considered validated for Sri Lankan Scotch Bonnet farms [4]. Units and calibration versions should be included with sensor results that have been calibrated. A specialist in agriculture should version and review crop spacing and suitability thresholds.

Additionally, requirements, design choices, implementation evidence, and research findings must be clearly separated in professional practice. Without supporting documentation, such as code, test logs, evaluation data, or user research records, the report shouldn't claim that a function is implemented, tested, or effective. Assumptions and uncertainty should be disclosed in financial predictions. The user should be able to adjust inputs and maintain control over the projected plant count. High-impact

recommendations should include a path to manual or expert review along with an explanation of their reasoning. [3]

Lastly, rather than being added after deployment, security and privacy restrictions should be included into the system. Farm location, photos, and financial data should only be accessible for approved uses. Procedures for incident handling, report sharing, retention, and deletion should be recorded. Failure modes that can result in excessive investment, risky land clearing, or improper crop-care practices should be evaluated. These procedures encourage responsible decision-making and responsibility. [3]

RFC 7946, which includes WGS 84 longitude–latitude coordinate order, is followed in GeoJSON interchange [13]. Planar meters do not apply to geographic coordinates. The service maintains valid polygon rings and records its geodesic or projected-area approach. The map interface needs maintained polygon controls instead of the retired dependence because the Google Maps Drawing Library is no longer available as of May 2026 [14].

The implementation and measurable outcomes should be reported in the subsequent thesis stage in accordance with the specifications and evaluation procedure outlined in these chapters.

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APPENDIX A: REQUIREMENTS TRACEABILITY MATRIX

This appendix links each functional requirement in Table 4.1 to the objective it serves (Section 1.5.2), the service layer that realises it (Table 4.3) and the subsystem evaluation measures that would test it (Table 3.2). The non-functional requirements of Table 4.2 are linked to the sections of the thesis in which they are addressed. The appendix summarises relationships that are already stated in the thesis and does not introduce new requirements.

Table A.1 Traceability of functional requirements

Requirement	Objective (Section 1.5.2)	Service layer (Table 4.3)	Evaluation subsystem (Table 3.2)
FR-01 Plot input	Objective 1: land-analysis workflow	1 User input and plot selection	Area and capacity estimation
FR-02 Land-cover analysis	Objective 1: land-analysis workflow	2 Imagery acquisition and preprocessing 3 Model 1 land-cover service	Land-cover segmentation
FR-03 Preliminary capacity	Objective 1: land-analysis workflow	3 Model 1 land-cover service	Area and capacity estimation
FR-04 Ground-photo analysis	Objective 2: two-photo preparation workflow	4 Model 2 preparation service	Ground-photo model
FR-05 Farmer confirmation	Objective 2: two-photo preparation workflow	4 Model 2 preparation service	Farmer confirmation
FR-06 IoT ingestion	Objective 3: IoT evidence pipeline	5 IoT and soil-evidence service	IoT pipeline
FR-07 Suitability evaluation	Objective 4: staged suitability decision process	6 Suitability decision engine	Suitability gate
FR-08 Resource inventory	Objective 5: resource limits and feasible plant count	7 Resource and cost engine	Integrated workflow
FR-09 Feasible plant count	Objective 5: resource limits and feasible plant count	7 Resource and cost engine 8 Feasibility and financial engine	Integrated workflow
FR-10 Financial estimation	Objective 6: financial estimation procedure	8 Feasibility and financial engine	Financial engine

Requirement	Objective (Section 1.5.2)	Service layer (Table 4.3)	Evaluation subsystem (Table 3.2)
FR-11 Alternative plans	Objective 5: resource limits and feasible plant count	8 Feasibility and financial engine	Integrated workflow
FR-12 Traceable report	Objective 7: evaluation, provenance, privacy and reproducibility	9 Report and audit layer	Integrated workflow

Table A.2 Location of non-functional requirements in the thesis

Requirement	Design implication (Table 4.2)	Addressed in
NFR-01 Transparency	Separate model output, IoT measurement, farmer input, sourced value, proxy and assumption.	Sections 4.3 and 4.4.7; Table B.3
NFR-02 Reliability	Validate input files, polygons, sensor payloads, units and decision-rule preconditions.	Sections 4.4.3 and 4.4.8; Table 4.6
NFR-03 Security and privacy	Use authentication, authorization, protected transport/storage, controlled sharing and retention rules.	Section 3.6
NFR-04 Usability	Use guided photo capture, clear units, concise explanations, accessible language and correction paths.	Sections 3.6 and 4.4.2
NFR-05 Reproducibility	Version models, thresholds, costs, source data, calibration and simulation parameters.	Section 3.7
NFR-06 Robustness	Handle missing, stale, contradictory and low-confidence evidence without fabricating values.	Sections 4.4.3 and 4.4.4; Table 4.6
NFR-07 Maintainability	Keep image models, evidence ingestion, suitability rules and financial calculations modular.	Section 4.2
NFR-08 Responsible recommendations	Escalate uncertain, hazardous or agronomically consequential decisions to manual/expert review.	Sections 3.6 and 4.4.4; Table 4.6

APPENDIX B: SUMMARY OF FORMULAS AND DECISION RULES

This appendix collects, for quick reference, the evaluation metrics, calculation rules and decision categories that are defined in the body of the thesis. Each entry names the section in which it is defined and explained. The rules are design specifications; the worked example in Figure 4.4 uses illustrative design inputs only and is not a result.

Table B.1 Formulas used in the evaluation and decision design

Item	Expression	Defined in
Precision	TP / (TP + FP)	Section 3.4.1
Recall	TP / (TP + FN)	Section 3.4.1
F1-score	2TP / (2TP + FP + FN)	Section 3.4.1
Intersection over union (IoU)	TP / (TP + FP + FN) for each class; mean IoU is the unweighted mean over the evaluated classes	Section 3.4.1
Empirical interval coverage	Percentage of observed outcomes falling inside the forecast P10–P90 interval (about 80% for a well-calibrated interval); reported together with interval width	Section 3.4.1
Land-cover class share	$P_c = 100\ A_c / A$, where $A_c = \Sigma\ w_i\ I(label_i = c)$; w_i is the geographic area represented by pixel i and A is the polygon area	Section 4.4.1
Ready planting capacity	$floor(A_{ready} \times e / (s_{row} \times s_{plant}))$; e is the layout efficiency and spacing is measured in metres	Section 4.4.1
Two-photo condition fusion	$p_c = 1 - (1 - p_{A,c})(1 - p_{B,c})$	Section 4.4.2
Feasible plant count	min(farmer-planned count, approved physical capacity, soil/method capacity, budget capacity, labour capacity, water capacity)	Section 4.4.5
First-season cash outflow	K + O, where K is new capital purchases and O is seasonal cash operating expenditure	Section 4.4.6.1
Seasonal economic cost	$C_e = O + L + D$, where L is imputed family labour and D is owned-asset usage and capital consumption	Section 4.4.6.1
Salable output	Q = planted count × survival fraction × yield per surviving plant × quality acceptance fraction	Section 4.4.6.1
Revenue	$R = \Sigma\ (Q_m \times P_m)$ over harvest months m	Section 4.4.6.1
Benefit–cost ratio	R / C_e	Section 4.4.6.1
Break-even price	C_e / Q	Section 4.4.6.1

Item	Expression	Defined in
Economic profit	$R - C_e$	Section 4.4.6.1
Economic ROI (%)	$100\ (R - C_e) / C_e$	Section 4.4.6.1
Break-even yield (fixed-cost simplification)	C_e / P	Section 4.4.6.1
Required financing	$\max(0, -\min_t CF_t)$, where CF_t is the cumulative net cash flow	Section 4.4.6.1
Illustrative ROI sensitivity (design example only)	ROI = 100 (1.25 × yield multiplier × price multiplier − 1)	Section 4.4.6.2; Figure 4.4

Table B.2 Suitability classes and the provisional status

Class or status	Meaning	Treatment in NAIMIRIS
S1	Highly suitable	A class is assigned only when the required evidence is complete. The most severe verified limiting factor determines the overall class, and the factor and its supporting data are recorded.
S2	Moderately suitable	As for S1.
S3	Marginally suitable	As for S1.
N1	Currently not suitable	As for S1. Potential suitability after documented improvements is reported separately from current suitability.
N2	Permanently not suitable	As for S1. N2 reflects the established land-use and evaluation assumptions, not a permanent ban on all possible uses (Section 4.4.4).
PROVISIONAL – NOT ASSIGNED	Required evidence is missing or stale	No final class is returned. The missing evidence, noted limitations and safe interim actions are listed. Image confidence alone never yields a final class.

Table B.3 Evidence categories kept separate in every report

Category	Typical content in this design	Treatment
Model output	Land-cover class estimates (Model 1) and visible preparation-condition probabilities (Model 2)	Reported with confidence and warnings; never used alone to assign a final suitability class.

Category	Typical content in this design	Treatment
IoT measurement	Timestamped soil moisture, soil temperature, air temperature and air relative humidity	Raw and calibrated values are kept; duplicate, stale or implausible readings are rejected or marked. They cannot establish soil texture, pH or EC.
Farmer input	Plot boundary, land size, budget, planned plants, workers, working hours, owned equipment, water source and confirmed affected-fraction bands	Initial entries are distinguished from later corrections; the farmer can review and modify them.
Sourced value	Official crop guidance, laboratory results, soil maps, and dated unit costs and prices	Recorded with source, date and unit so that the value can be audited.
Proxy	Crop-family (capsicum/chilli) guidance, or a price from another channel when no local Nai Miris price exists	Shown only when explicitly labelled, with a warning and reduced confidence.
Assumption	Layout efficiency, affected-fraction mapping, task-hour coefficients and simulation distributions	Versioned and displayed in the report so that users can review or modify them.

APPENDIX C: ACADEMIC INTEGRITY AND AI USE DISCLOSURE

This appendix records the generative AI and other automated tools used in preparing this thesis, in accordance with the module policy referred to in the Declaration. For each tool it states the purpose, the workflow followed, the verification performed and the sections or artefacts affected.

Table C.1 Record of AI and automated tool use

Tool and version	Purpose	Representative prompts or workflow	Verification performed	Sections or artefacts affected
OpenAI ChatGPT;	Assist with thesis drafting, organisation and English language improvement using supplied project documents.	Use my proposal, thesis draft and backend specification to develop the report according to the individual thesis guidelines, preserving citation numbers	Pending author verification: compare generated content with the original project documents and actual implementation; correct unsupported statements	Chapters 1–4 and supporting appendices.
Claude Ai	Assist with finding supporting literature and preparing reference information.	Find relevant research sources for land-cover analysis, ground-image analysis and agricultural decision support, and provide numbered references.	Pending author verification: open each original source, confirm bibliographic details and check that it supports the associated claim.	Literature review, supporting citations and reference list.

The candidate remains responsible for the accuracy, originality, citations, code, data, analyses and conclusions in this thesis.